



Faculty of Computer Applications

BCA Sem-1
(AY-2026-27)

Lab Book

Sub Code : 05BC3103
(Programming Practices : 1)

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GitHub Link | <https://github.com/zmeetrajsinh04/BCA-1-pp-1---92600527022--MEETRAJSINH--ZALA.git>

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4.	Write a program that input value from user and find out Area of Circle.
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6.	Write a program that input 3 values from user and find out its Average.
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8.	Write a program Interchange the value of two numbers using third variable.
9.	Write a program Interchange the value of two numbers without using third variable.
10.	Write a program that input quantity, price, discount and display final amount.

(1) Write a Program to Print "Hello World".

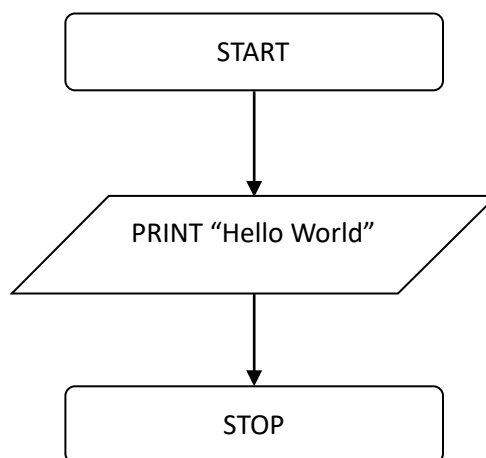
Algorithm :

Step 1 : START

Step 2 : PRINT "Hello World"

Step 3 : STOP

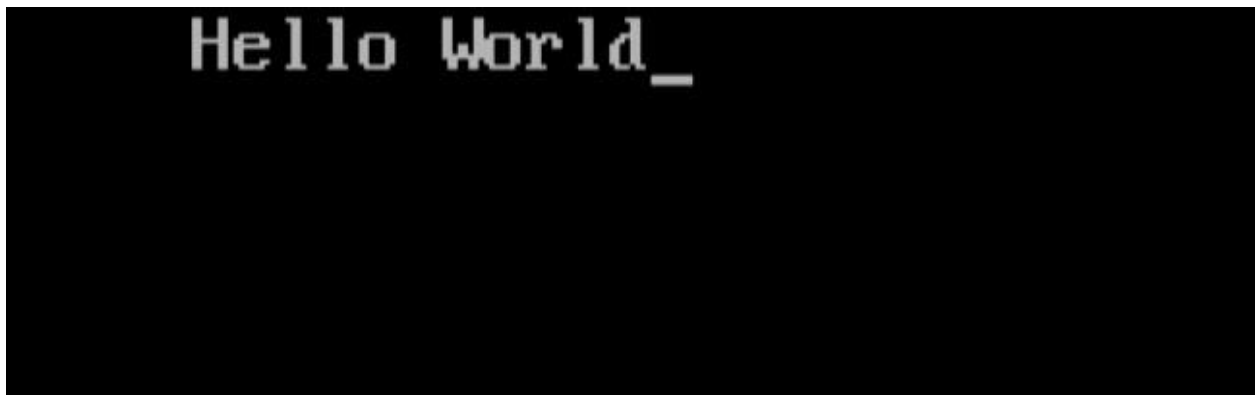
Flowchart :



)

```
void main(){      clrscr();  
printf("Hello World");  
    getch();  
}
```

Output :



(2 Write a Program that takes 2 Values from user and Perform All Arithmetic Operation.

Algorithm :

Step 1 : START
Step 2 : DECLARE variables x=0.0, y=0.0, a, s, m, d
Step 3 : INPUT x,y
Step 4 : a = x + y
Step 5 : s = x - y
Step 6 : m = x * y
Step 7 : d = x / y

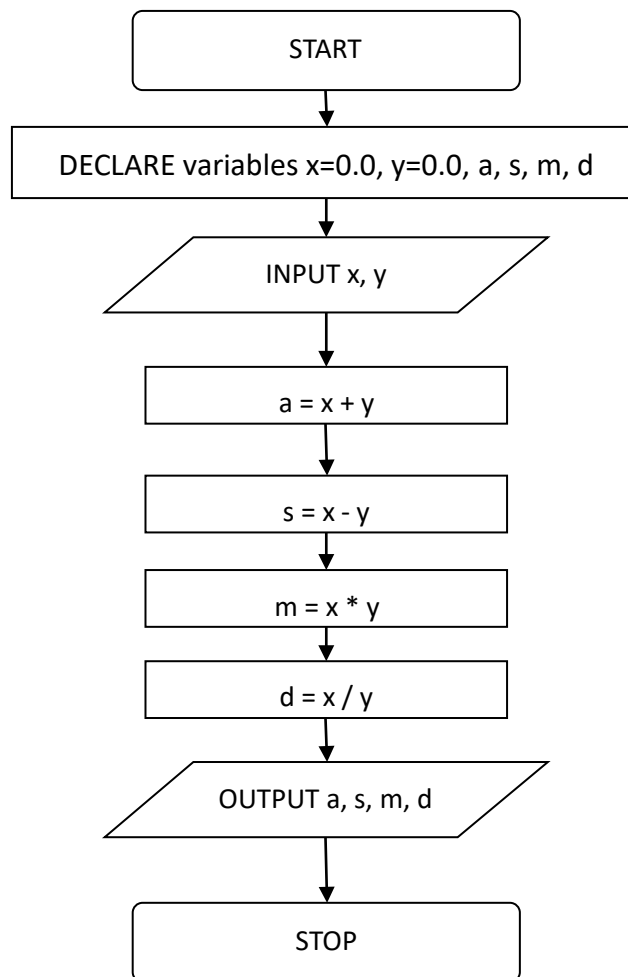
Code :

```
#include<stdio.h>
#include<conio.h>
```

Step 8 : OUTPUT a, s, m, d

Step 9 : STOP

Flowchart :



```
void main()
{
    float x,y;
    clrscr();
```

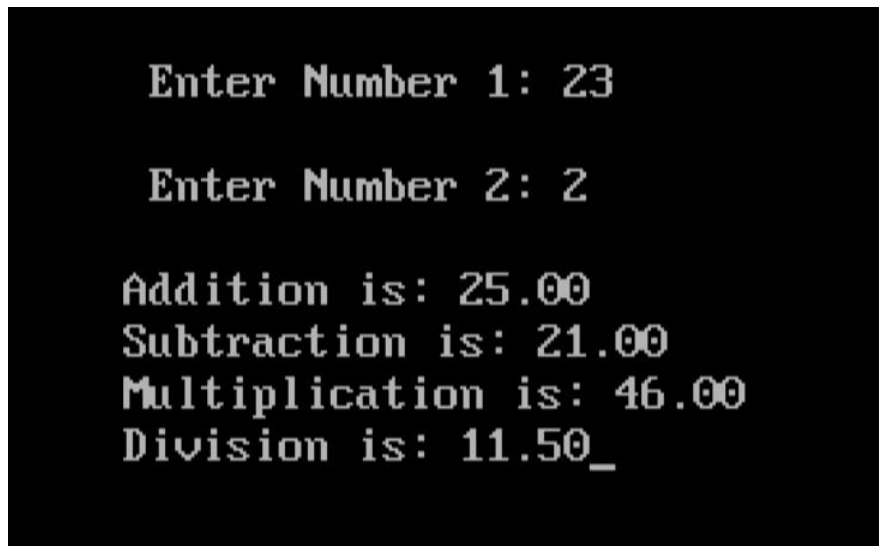
)

```
printf("\n Enter Number 1: "); scanf("%f",&x);  
printf("\n Enter Number 2: "); scanf("%f",&y);
```

```
printf("Addition Is: %.2f \n",x+y);  
printf("Subtraction Is: %.2f \n",x-y);  
printf("Multiplication Is: %.2f \n",x*y);  
printf("Division Is: %.2f \n",x/y);  
  
getch();
```

```
}
```

Output :

A screenshot of a terminal window with a black background and white text. The text shows the program's output for the given inputs. It starts with 'Enter Number 1: 23' and 'Enter Number 2: 2'. Then it displays the results of four operations: 'Addition is: 25.00', 'Subtraction is: 21.00', 'Multiplication is: 46.00', and 'Division is: 11.50_'.

```
Enter Number 1: 23  
  
Enter Number 2: 2  
  
Addition is: 25.00  
Subtraction is: 21.00  
Multiplication is: 46.00  
Division is: 11.50_
```

(3 Write a program that takes 3 values for Principle Amount, Rate of Interest and No. of Years and find out Simple Interest.

Algorithm :

Step 1 : START

Step 2 : DECLARE variables SI, P, R, T

Code :

```
#include<stdio.h>
#include<conio.h>
```

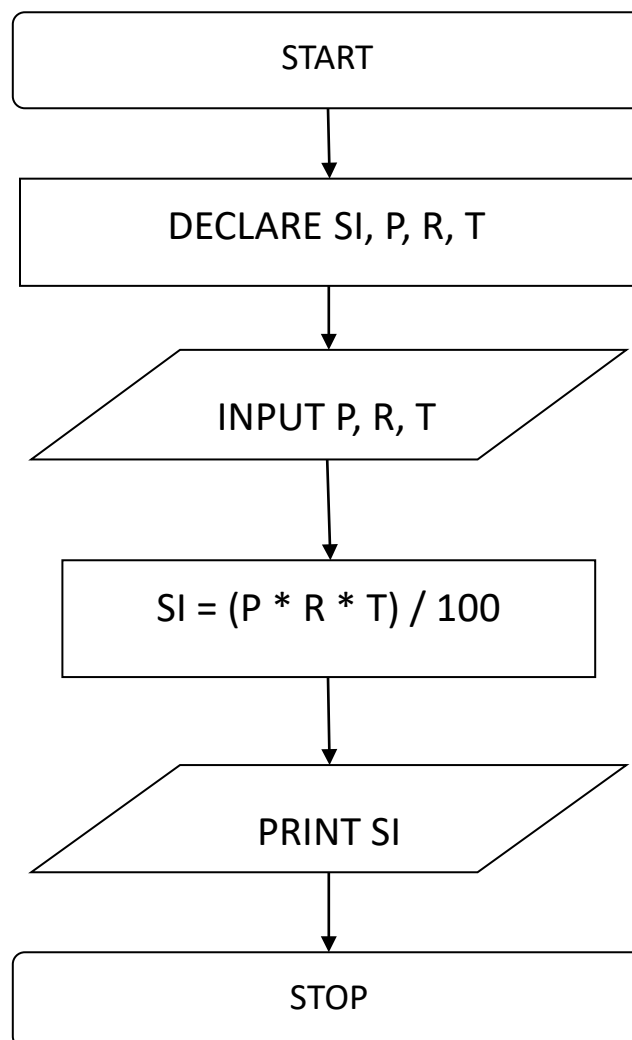
Step 3 : INPUT P, R, T

Step 4 : $SI = (P * R * T) / 100$

Step 5 : PRINT SI

Step 6 : STOP

Flowchart :

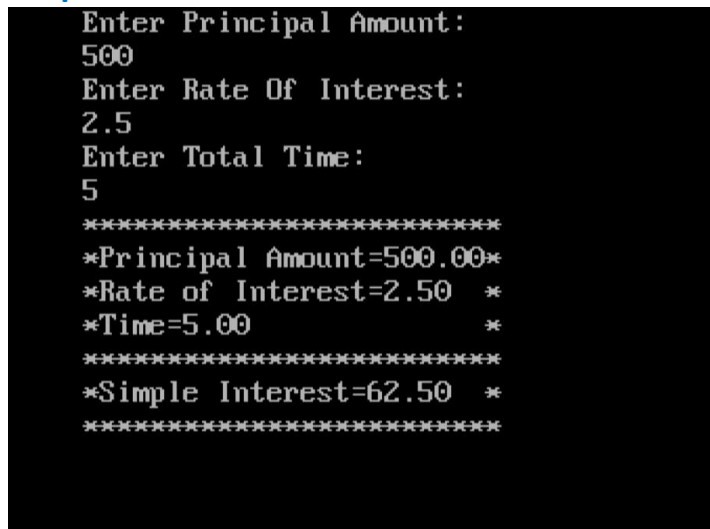


)

```
void main(){
    float SI,P,R,T;      clrscr();
    printf("Enter Principal Amount: \n");
    scanf("%f",&P);
    printf("Enter Rate Of Interest: \n");
    scanf("%f",&R);
    printf("Enter Total Time: \n");
    scanf("%f",&T);    SI=(P*R*T)/100;
        printf("*****\n");
        printf("*Principal Amount=%.2f*\n",P);
        printf("*Rate of Interest=%.2f *\n",R);
        printf("*Time=%.2f *\n",T);
        printf("*****\n");
        printf("*Simple Interest=%.2f *\n",SI);
        printf("*****");

        getch();
}
```

Output :



```
Enter Principal Amount:
500
Enter Rate Of Interest:
2.5
Enter Total Time:
5
*****
*Principal Amount=500.00*
*Rate of Interest=2.50 *
*Time=5.00 *
*****
*Simple Interest=62.50 *
*****
```

(4 Write a program that input value from user and find out Area of Circle.

Code :

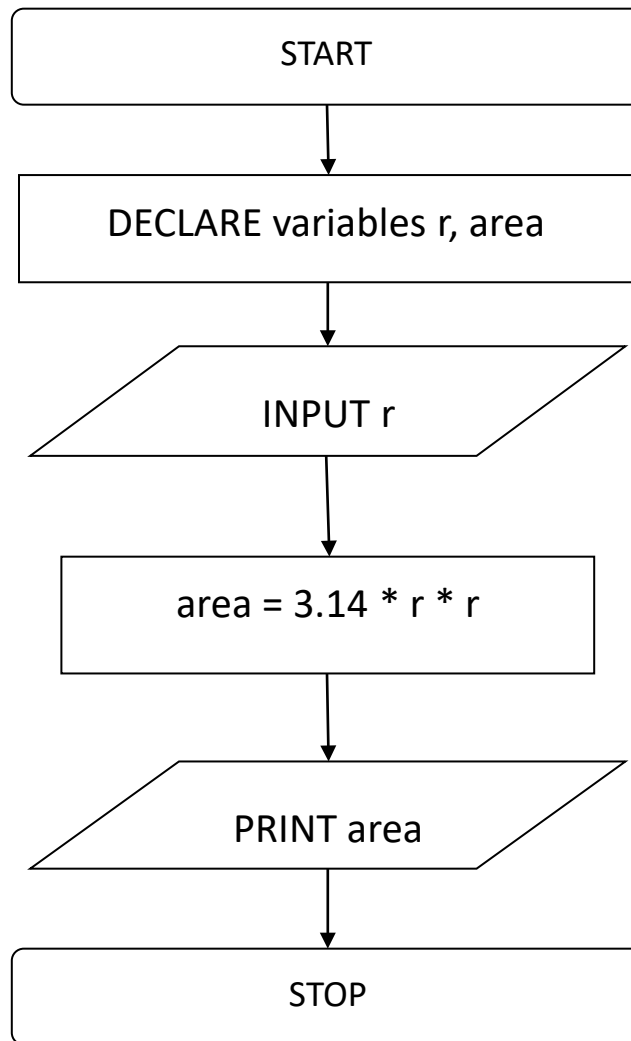
```
#include<stdio.h>
#include<conio.h>
```

Algorithm :

Step 1 : START
Step 2 : DECLARE variable r, area
Step 3 : INPUT r
Step 4 : $\text{area} = 3.14 * r * r$
Step 5 : PRINT area
Step 6 : STOP

Flowchart :

)



```
void main()
{
    float Radius;
    clrscr();
    printf("Enter Radius Of Circle: \n");
    scanf("%f",&Radius);
    printf("The Area Of Circle is: %f",3.14*Radius*Radius);
    getch();
}
```

Code :

```
#include<stdio.h>
#include<conio.h>
```

Output :

A screenshot of a terminal window with a black background and white text. The text displayed is: "Enter Radius Of Circle:" followed by the input "3" on the next line, and then "The Area Of Circle is: 28.260000_" on the third line.

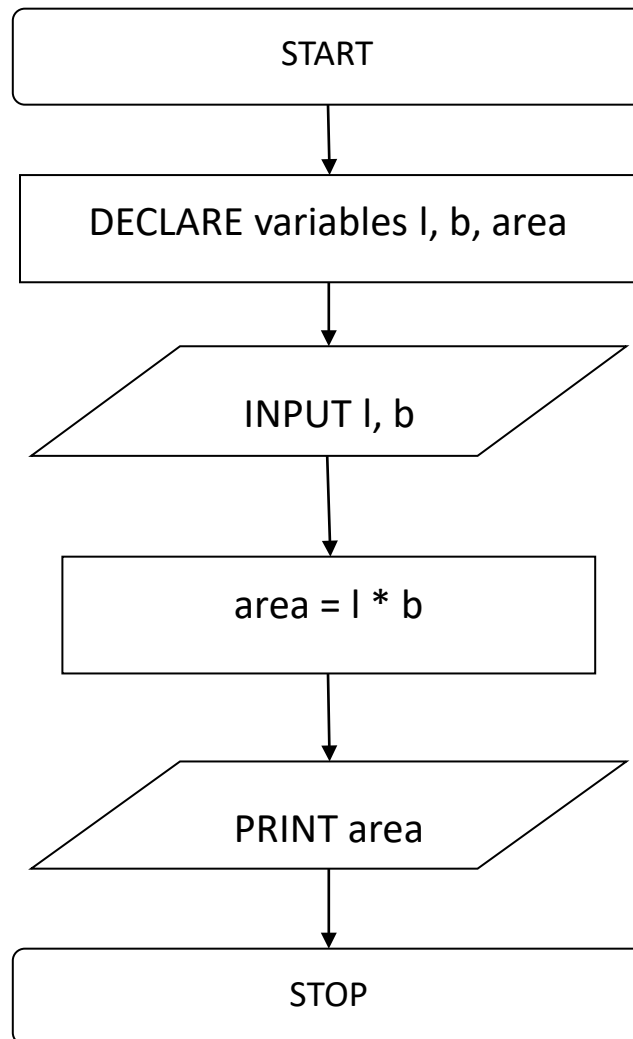
(5 Write a program that input two values from user and find out Area of Rectangle.

Algorithm :

Step 1 : START
Step 2 : DECLARE variable l, b, area
Step 3 : INPUT l, b
Step 4 : $\text{area} = l * b$
Step 5 : PRINT area
Step 6 : STOP

Flowchart :

)

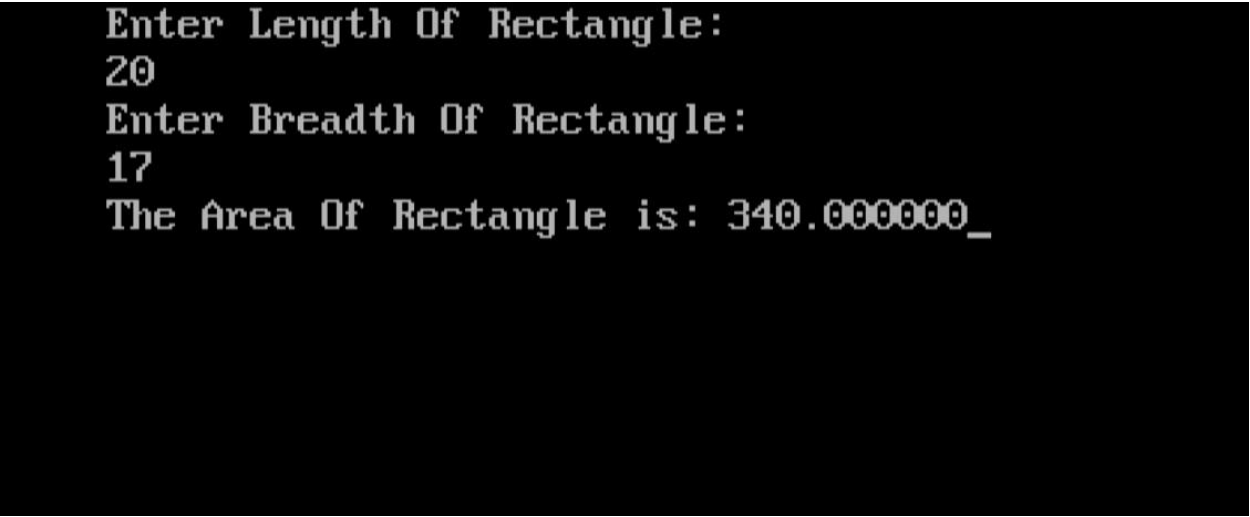


```
void main()
{
    float Length,Breadth;    clrscr();
    printf("Enter Length Of Rectangle: \n");
    scanf("%f",&Length);
    printf("Enter Breadth Of Rectangle: \n");
    scanf("%f",&Breadth);
    printf("The Area Of Rectangle is: %f",Length*Breadth);
    getch();
}
```

Code :

```
#include<stdio.h>
#include<conio.h>
```

Output :



```
Enter Length Of Rectangle:
20
Enter Breadth Of Rectangle:
17
The Area Of Rectangle is: 340.000000_
```

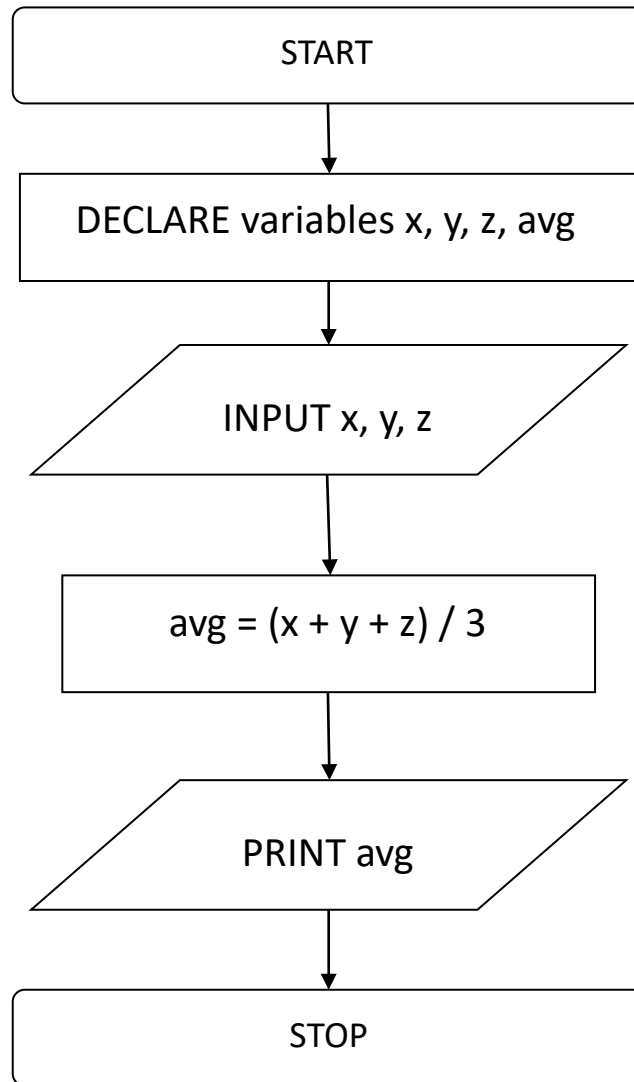
(6 Write a program that input 3 values from user and find out its Average.

Algorithm :

```
Step 1 : START
Step 2 : DECLARE variables x, y, z, avg
Step 3 : INPUT x, y, z
Step 4 : avg = (x + y + z) / 3
Step 5 : PRINT avg
Step 6 : STOP
```

)

Flowchart :



Code :

```
#include<stdio.h>
#include<conio.h>

void main()
{
    int x,y,z;
    clrscr();
    printf("\n Enter Number 1: "); scanf("%d",&x);
    printf("\n Enter Number 2: "); scanf("%d",&y);
    printf("\n Enter Number 3: ");
    scanf("%d",&z);
    printf("\n Average is: %d ",(x+y+z)/3);
    getch();
}
```

Output :

)

```
Enter Number 1: 23
```

```
Enter Number 2: 33
```

```
Enter Number 3: 43
```

```
Average is: 33 _
```

(7 Write a program to Calculate the Square and Cube of a number.

Algorithm :

Step 1 : START

Step 2 : DECLARE variables n, sqr, cube

Step 3 : INPUT n

Step 4 : $sqr = n * n$

Step 5 : $cube = n * n * n$

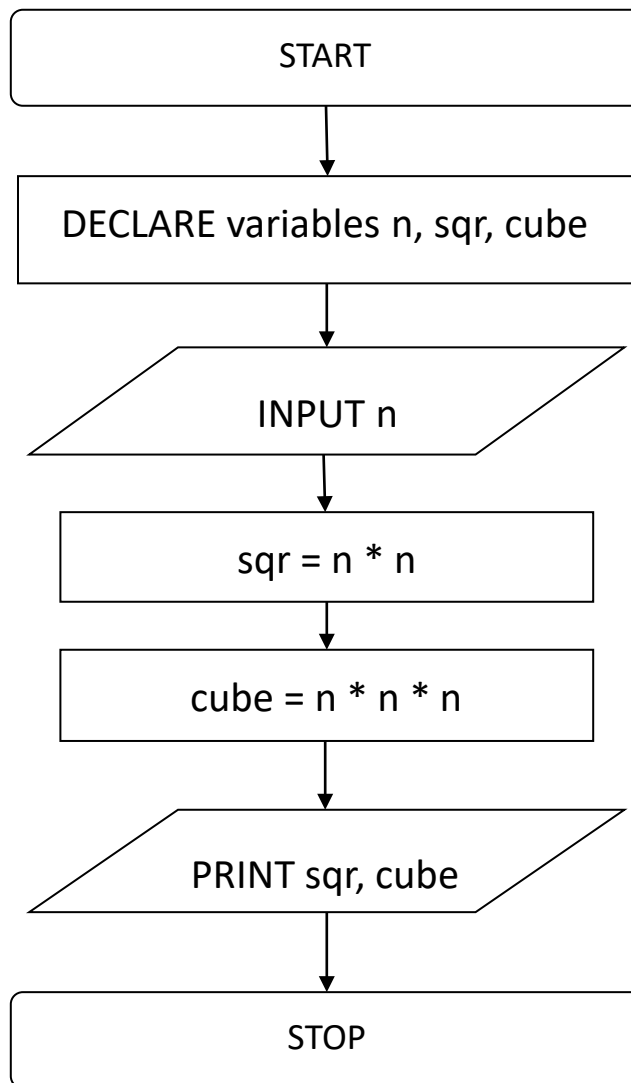
Step 6 : OUTPUT sqr, cube

Step 7 : STOP

Code :

```
#include<stdio.h>
#include<conio.h>
```

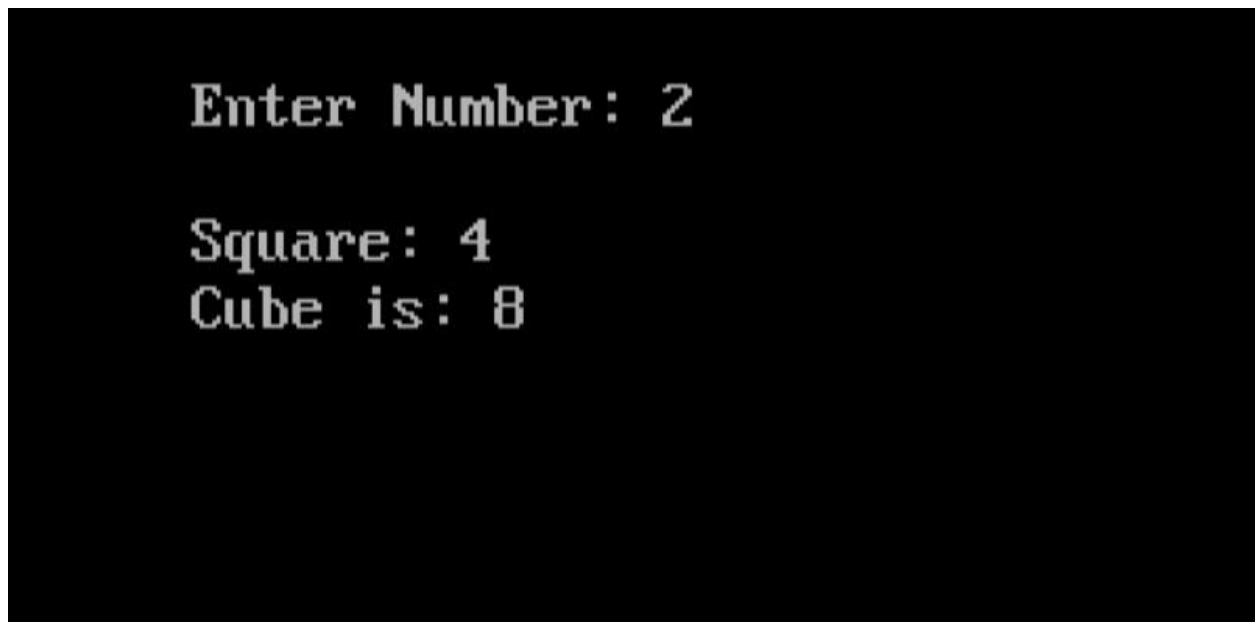
Flowchart :



```
void main()
{
    int n;
    clrscr();
    printf("\n Enter Number: ");    scanf("%d",&n);
```

```
)  
  
    printf("\n Square is: %d",n*n); printf("\n Cube is:  
%d",n*n*n);  
        getch();  
  
}
```

Output :

A screenshot of a terminal window with a black background and white text. The text shows the output of a C program. It starts with 'Enter Number: 2', followed by a blank line, then 'Square: 4', and finally 'Cube is: 8'.

(8 Write a program Interchange the value of two numbers using third variable.

Algorithm :

Step 1 : START

Code :

```
#include<stdio.h>  
#include<conio.h>
```

Step 2 : DECLARE variables temp, x, y

Step 3 : INPUT x, y

Step 4 : temp = x

Step 5 : x = y

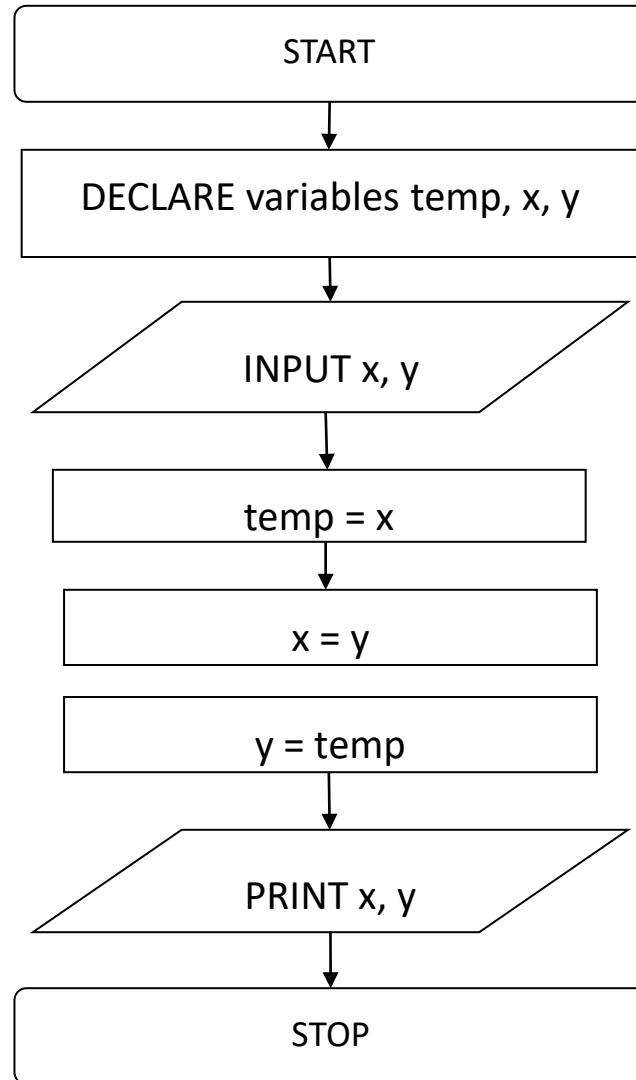
Step 6 : y = temp

Step 7 : PRINT x, y

Step 8 : STOP

)

Flowchart :



```
void main()  
{
```

```
    int temp,x,y;  
    clrscr();  
    printf("\n Enter Number 1: ");  
    scanf("%d",&x);  
    printf("\n Enter Number 2: ");
```

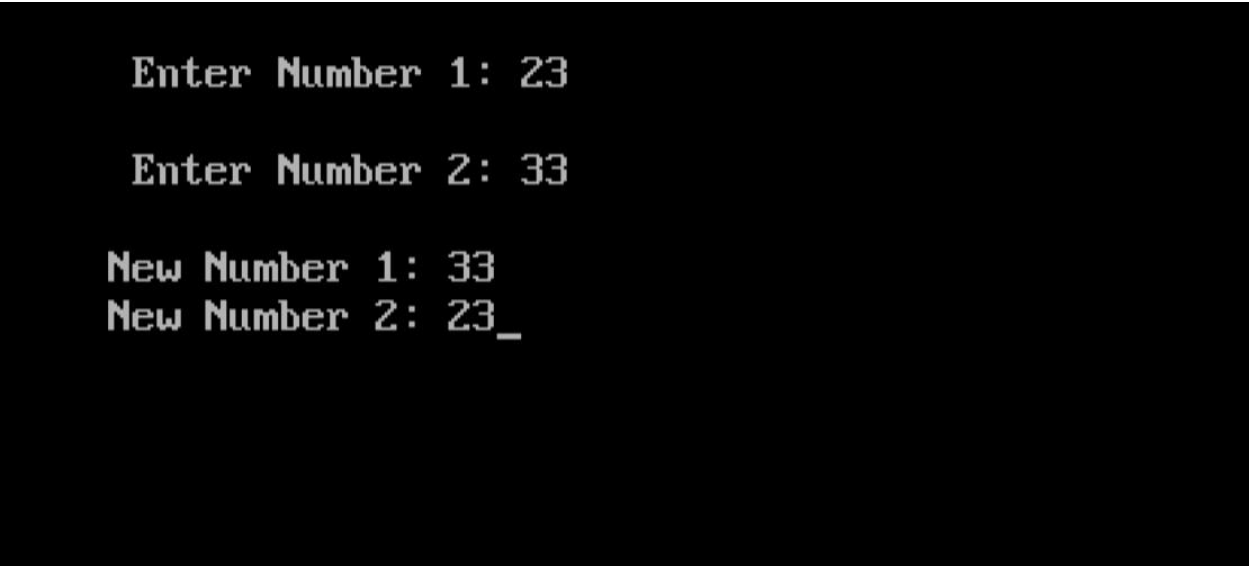
Code :

```
#include<stdio.h>
#include<conio.h>

scanf("%d",&y);
temp=x;    x=y;
           y=temp;
printf("\nNew Number 1: %d",x);
printf("\nNew Number 2: %d",y);    getch();

}
```

Output :

A screenshot of a terminal window with a black background and white text. It shows the output of a C program. The first two lines are prompts to enter numbers: "Enter Number 1: 23" and "Enter Number 2: 33". The next two lines show the swapped values: "New Number 1: 33" and "New Number 2: 23_".

```
Enter Number 1: 23

Enter Number 2: 33

New Number 1: 33
New Number 2: 23_
```

(9 Write a program Interchange the value of two numbers without using third variable.

Algorithm :

Step 1 : START

)

Step 2 : DECLARE variables x, y

Step 3 : INPUT x, y

Step 4 : $x = x + y$

Step 5 : $y = x - y$

Step 6 : $x = x - y$

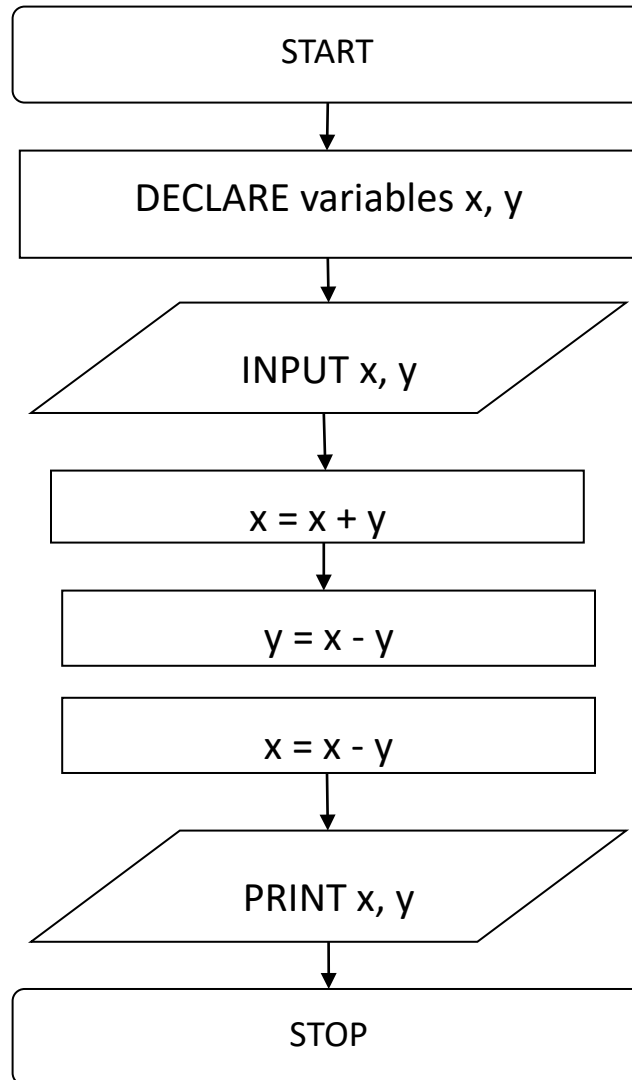
Step 7 : PRINT x, y

Step 8 : STOP

Flowchart :

Code :

```
#include<stdio.h>
#include<conio.h>
```

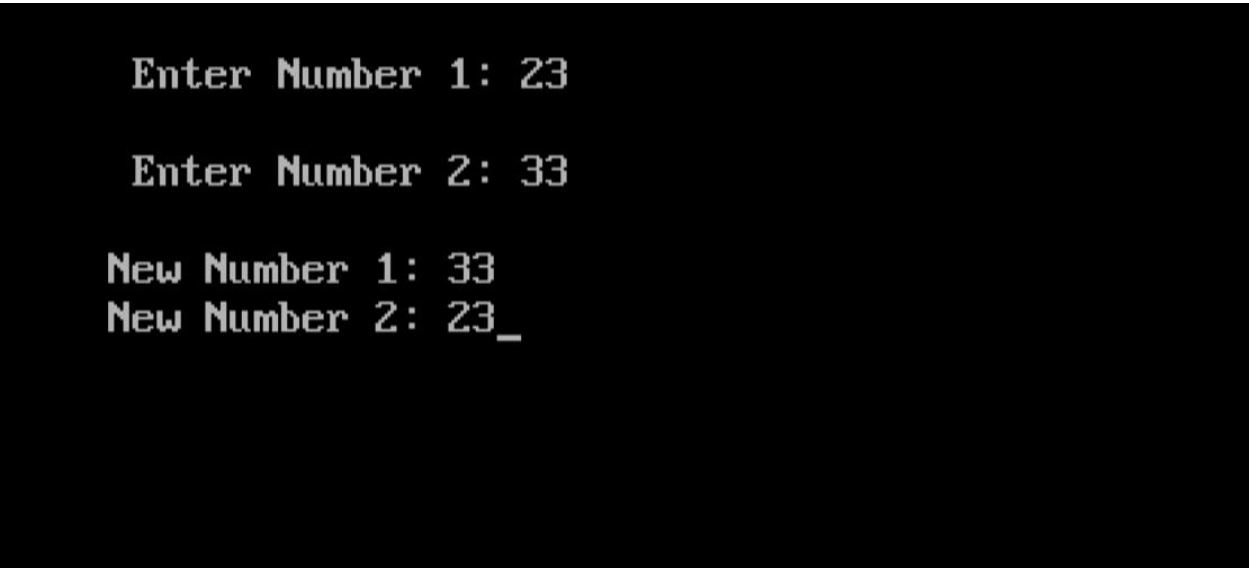


```
void main()
{
    int x,y;    clrscr();
                printf("\n Enter Number 1: ");
                scanf("%d",&x);
                printf("\n Enter Number 2: ");
```



```
)  
  
    scanf("%d",&y);  
x=x+y;    y=x-y;    x=x-y;  
    printf("\nNew Number 1: %d",x);  
printf("\nNew Number 2: %d",y);    getch();  
  
}
```

Output :

A screenshot of a terminal window with a black background and white text. It shows the output of a C program. The first two lines prompt the user to enter two numbers: 'Enter Number 1: 23' and 'Enter Number 2: 33'. The next two lines show the calculated results: 'New Number 1: 33' and 'New Number 2: 23_'.

```
Enter Number 1: 23  
  
Enter Number 2: 33  
  
New Number 1: 33  
New Number 2: 23_
```

Code :

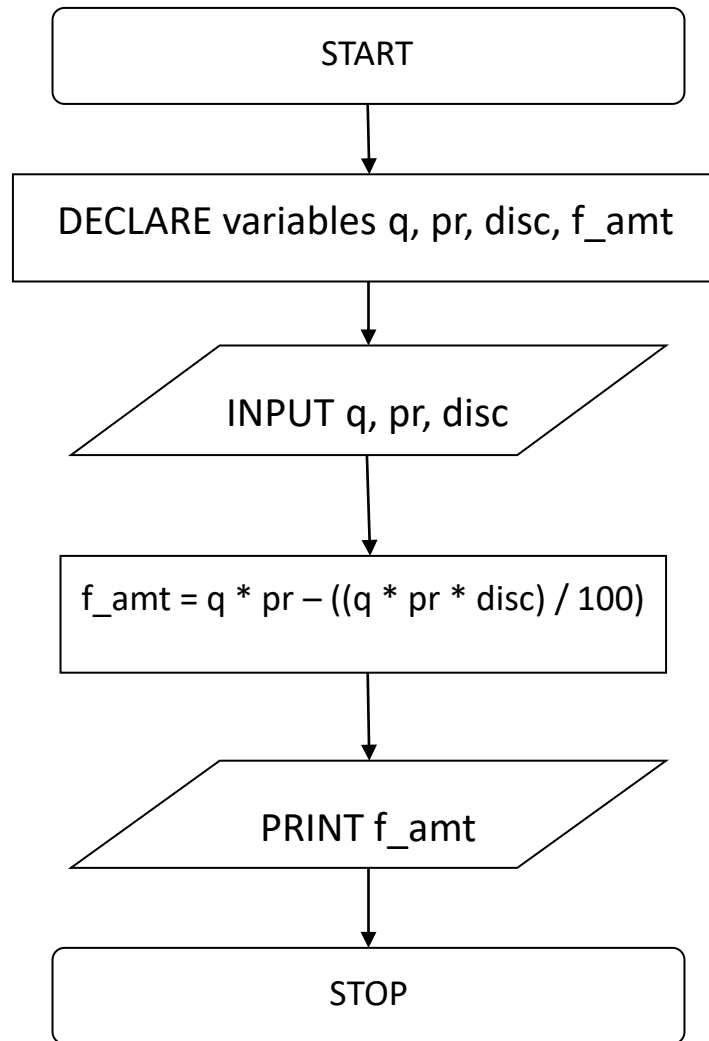
```
#include<stdio.h>
#include<conio.h>
```

(10) Write a program that input quantity, price, discount and display final amount.

Algorithm :

Step 1 : START
Step 2 : DECLARE variables q, pr, disc, f_amt
Step 3 : INPUT q, pr, disc
Step 4 : $f_amt = q * pr - ((q * pr * disc) / 100)$
Step 5 : OUTPUT f_amt
Step 6 : STOP

Flowchart :



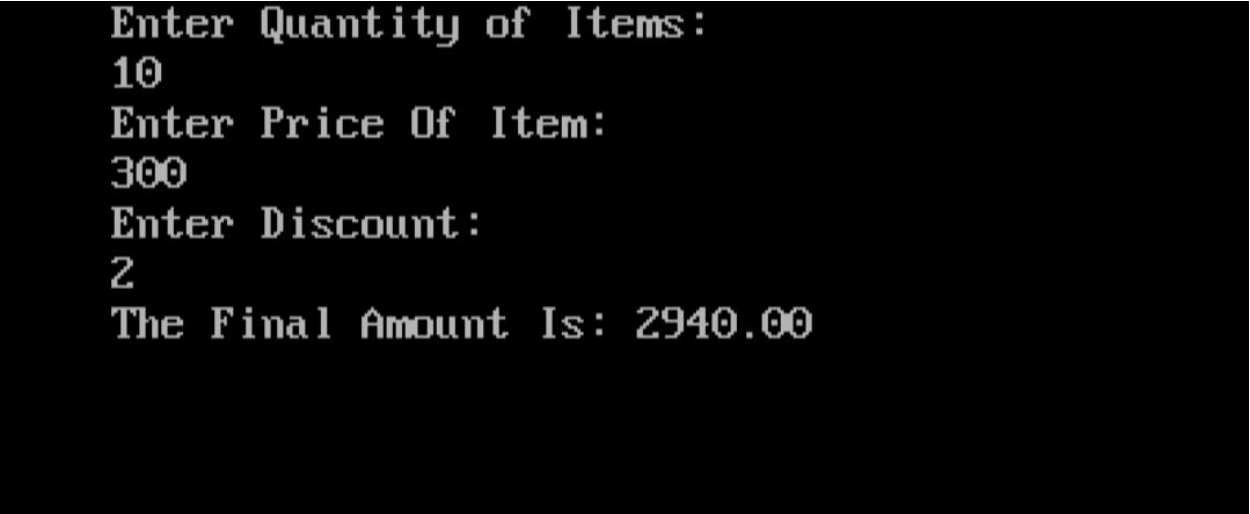
```
void main(){  
  
    float Input_quantity,Price,Discount,Final_Amount;  
    clrscr();  
    printf("Enter Quantity of Items: \n");  
    scanf("%f",&Input_quantity);  
    printf("Enter Price Of Item: \n");
```

Code :

```
#include<stdio.h>
#include<conio.h>

scanf("%f",&Price);          printf("Enter
Discount:                    \n");
scanf("%f",&Discount);
    Final_Amount = Input_quantity*Price-((Input_quantity*Price*Discount)/100);
printf("The Final Amount Is: %.2f",Final_Amount);    getch();
}
```

Output :

A screenshot of a terminal window with a black background and white text. The text shows the program's execution: it prompts for the quantity of items (10), the price of an item (300), and the discount (2), then calculates and displays the final amount as 2940.00.

```
Enter Quantity of Items:
10
Enter Price Of Item:
300
Enter Discount:
2
The Final Amount Is: 2940.00
```